

Recall from last class:

- Define  $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$  by  $T(\vec{x}) = A\vec{x}$  where  $A$  is an  $m$  by  $n$  matrix. Recall we call  $\mathbb{R}^n$  the domain and  $\mathbb{R}^m$  the codomain. The set of outputs,  $\{T(\vec{x}) : \vec{x} \in \mathbb{R}^n\}$ , is the image or the range of  $T$ .
- $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$  is **linear** if
  1.  $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$  for all  $\vec{u}, \vec{v}$  in  $\mathbb{R}^n$ ,
  2.  $T(c\vec{u}) = cT(\vec{u})$  for all  $\vec{u}$  in  $\mathbb{R}^n$  and  $c$  in  $\mathbb{R}$ .
- **Properties of Linear Transformations.**
  1. If  $T$  is linear, then  $T(\vec{0}) = \vec{0}$ .
  2. If  $T$  is linear, then  $T(c\vec{u} + d\vec{v}) = cT(\vec{u}) + dT(\vec{v})$ .
- All matrix transformations are linear.
- If  $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$  is given by  $T(\vec{x}) = A\vec{x}$ , then the image of  $T$  is the span of the columns of  $A$ ; namely,
 
$$\text{Im}(T) = \text{Span}\{\vec{a}_1, \dots, \vec{a}_n\}.$$
- We also saw various geometric transformations, including projections, dilations, and shears.

1. Suppose  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$  is linear and

$$T \left( \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 5 \\ -7 \\ 2 \end{bmatrix}, \text{ and } T \left( \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} -3 \\ 8 \\ 0 \end{bmatrix}.$$

What is

$$T \left( \begin{bmatrix} a \\ b \end{bmatrix} \right)?$$

**Theorem 10:** Let  $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$  be linear. Then there exists a unique matrix  $A$  such that  $T(\vec{x}) = A\vec{x}$  for all  $\vec{x} \in \mathbb{R}^n$ .

2. Prove the Theorem.

3. Let  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  be the linear transformation given by rotating  $\vec{x}$  by  $\theta$  counterclockwise. Find the associated matrix of  $T$ .

4. Let  $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$  be defined by  $T(\vec{x}) = A\vec{x}$  where

$$A = \begin{bmatrix} 1 & -4 & 8 & 1 \\ 0 & 2 & -1 & 3 \\ 0 & 0 & 0 & 5 \end{bmatrix}.$$

Is  $T$  injective and/or surjective?

**Theorem 11:** Let  $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$  be linear. Then  $T$  is injective if and only if  $T(\vec{x}) = \vec{0}$  has only the trivial solution.

5. Prove the theorem.

**Theorem 12:** Let  $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$  be linear, and let  $A$  be the associated matrix. Then

- (a)  $T$  is surjective if and only if the columns of  $A$  span  $\mathbb{R}^m$ .
- (b)  $T$  is injective if and only if the columns of  $A$  are linearly independent.

6. Prove the theorem.

# Linear and Matrix Transformations – Answers / Solutions

1. Using linearity, we get

$$\begin{aligned} T\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) &= T\left(a\begin{bmatrix} 1 \\ 0 \end{bmatrix} + b\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) \\ &= aT\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) + bT\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) \\ &= a\begin{bmatrix} 1 \\ 0 \end{bmatrix} + b\begin{bmatrix} -3 \\ 8 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 5 & -3 \\ -7 & 8 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}. \end{aligned}$$

Thus  $T$  is also a matrix transformation. A natural question is whether every linear transformation is a matrix transformation.

2. We proceed as in the last example. First Let  $\vec{e}_j$  be the vector in  $\mathbb{R}^n$  with 1 in the  $j$ -th position and 0 elsewhere. Let  $x \in \mathbb{R}^n$ , then

$$\vec{x} = x_1\vec{e}_1 + \cdots + x_n\vec{e}_n.$$

Thus using the linearity of  $T$ , we get

$$\begin{aligned} T(\vec{x}) &= x_1T(\vec{e}_1) + \cdots + x_nT(\vec{e}_n) \\ &= [T(\vec{e}_1) \ \cdots \ T(\vec{e}_n)] \vec{x}. \end{aligned}$$

Thus  $T(\vec{x}) = A\vec{x}$  where  $A = [T(\vec{e}_1) \ \cdots \ T(\vec{e}_n)]$ .

In order to prove uniqueness, we suppose there is another matrix  $B$  such that  $T(\vec{x}) = B\vec{x}$  for all  $x \in \mathbb{R}^n$ . We want to show that  $A = B$ . Let  $B = [\vec{b}_1 \ \cdots \ \vec{b}_n]$ . Since  $A\vec{e}_j = B\vec{e}_j$ , we get  $\vec{b}_j = T(\vec{e}_j)$ . Thus  $A$  and  $B$  have the same columns, and thus  $A = B$  and  $A$  is unique.

3. Rotating  $\vec{e}_1$  and  $\vec{e}_2$  by  $\theta$  gives the associated matrix of  $T$  is  $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ .
4. We have a pivot position in each row, so  $T$  is onto (surjective).  $A\vec{x} = \vec{0}$  has free variables, so  $T$  is not one-to-one (injective).
5. Let  $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$  be linear.

$\Rightarrow$  Suppose  $T$  is one-to-one (injective). Since  $T$  is linear, we know  $\vec{x} = \vec{0}$  is a solution. Since  $T$  is injective, we know  $T(\vec{x}) = \vec{0}$  has at most one solution. Thus  $T(\vec{x}) = \vec{0}$  has only the trivial solution.

$\Leftarrow$  Suppose  $T(\vec{x}) = \vec{0}$  has only the trivial solution and  $T(\vec{u}) = T(\vec{v})$ . Since  $T$  is linear, we get  $T(\vec{u} - \vec{v}) = \vec{0}$ . Since  $T(\vec{x}) = \vec{0}$  has only the trivial solution, we know that  $\vec{u} - \vec{v} = \vec{0}$ , and hence  $\vec{u} = \vec{v}$ . Therefore  $T$  is one-to-one or injective.

6. Let  $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$  be linear and let  $A$  be the associated matrix.

**Proof of (a):**  $T$  is surjective means that for all  $\vec{b} \in \mathbb{R}^m$  there exists  $\vec{x} \in \mathbb{R}^n$ , or equivalently  $A\vec{x} = \vec{b}$  is consistent. By Theorem 4,  $A\vec{x} = \vec{b}$  is consistent for all  $\vec{b}$  if and only if the columns of  $A$  span  $\mathbb{R}^m$ .

**Proof of (b):** By Theorem 11,  $T$  is one-to-one if and only if  $T(\vec{x}) = \vec{0}$  has only the trivial solution. Thus the columns of  $A$  must be linearly independent.